

# Chi-Square Test for Categorical Variables

## Introduction

The chi-square test is a statistical method used to determine if there is a significant association between two categorical variables. This test is widely used in various fields, including social sciences, marketing, and healthcare, to analyze survey data, experimental results, and observational studies.

## Concept

The chi-square test is a non-parametric statistical method used to examine the association between two categorical variables. It evaluates whether the frequencies of observed outcomes significantly deviate from expected frequencies, assuming the variables are independent. The test is grounded in the chi-square distribution, which is applied to count data and helps in determining if any observed deviations could have arisen by random chance.

## Null Hypothesis and Alternative Hypothesis

The chi-square test involves formulating two hypotheses:

Null Hypothesis ( $H_0$ )(H0) - Assumes that there is no association between the categorical variables, implying that any observed differences are due to random chance.

Alternative Hypothesis ( $H_1$ )(H1) - Assumes that there is a significant association between the variables, indicating that the observed differences are not due to chance alone.

## Formula

The chi-square statistic is calculated using the formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where

$O_i$  is the observed frequency for category  $i$ .

$E_i$  is the expected frequency for category  $i$ , calculated as:

$$E_i = \frac{(\text{row total} \times \text{column total})}{\text{grand total}}$$

The sum is taken over all cells in the contingency table.

The calculated chi-square statistic is then compared to a critical value from the chi-square distribution table. This table provides critical values for different degrees of freedom ( $df$ ) and significance levels ( $\alpha$ ).

The degrees of freedom for the test are calculated as:

$$df = (r - 1) \times (c - 1)$$

where  $r$  is the number of rows and  $c$  is the number of columns in the table.

## Applications

- Market Research:** Analyzing the association between customer demographics and product preferences.
- Healthcare:** Studying the relationship between patient characteristics and disease incidence.
- Social Sciences:** Investigating the link between social factors (e.g., education level) and behavioral outcomes (e.g., voting patterns).
- Education:** Examining the connection between teaching methods and student performance.
- Quality Control:** Assessing the association between manufacturing conditions and product defects.

## Practical Example - Weak Correlation

Suppose a researcher wants to determine if there is an association between gender (male, female) and preference for a new product (like, dislike). The researcher surveys 100 people and records the following data:

| Category | Like | Dislike | Total |
|----------|------|---------|-------|
| Male     | 20   | 30      | 50    |
| Female   | 25   | 25      | 50    |
| Total    | 45   | 55      | 100   |

#### Step 1: Calculate Expected Frequencies

Using the formula for expected frequencies:

$$E_{Male, Like} = \frac{(50 \times 45)}{100} = 22.5 \quad E_{Male, Like} = 100(50 \times 45) = 22.5$$

$$E_{Male, Dislike} = \frac{(50 \times 55)}{100} = 27.5 \quad E_{Male, Dislike} = 100(50 \times 55) = 27.5$$

$$E_{Female, Like} = \frac{(50 \times 45)}{100} = 22.5 \quad E_{Female, Like} = 100(50 \times 45) = 22.5$$

$$E_{Female, Dislike} = \frac{(50 \times 55)}{100} = 27.5 \quad E_{Female, Dislike} = 100(50 \times 55) = 27.5$$

#### Step 2: Compute Chi-Square Statistic

$$\chi^2 = \frac{(20 - 22.5)^2}{22.5} + \frac{(30 - 27.5)^2}{27.5} + \frac{(25 - 22.5)^2}{22.5} + \frac{(25 - 27.5)^2}{27.5}$$

$$\chi^2 = 22.5(20 - 22.5)^2 + 27.5(30 - 27.5)^2 + 22.5(25 - 22.5)^2 + 27.5(25 - 27.5)^2$$

$$\chi^2 = \frac{(2.5)^2}{22.5} + \frac{(2.5)^2}{27.5} + \frac{(2.5)^2}{22.5} + \frac{(2.5)^2}{27.5}$$

$$\chi^2 = 22.5(2.5)^2 + 27.5(2.5)^2 + 22.5(2.5)^2 + 27.5(2.5)^2$$

$$\chi^2 = \frac{6.25}{22.5} + \frac{6.25}{27.5} + \frac{6.25}{22.5} + \frac{6.25}{27.5}$$

$$\chi^2 = 22.56.25 + 27.56.25 + 22.56.25 + 27.56.25$$

$$\chi^2 = 0.277 + 0.227 + 0.277 + 0.227$$

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$$\chi^2 = 1.008$$

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#### Step 3: Determine Degrees of Freedom

$$df = (2 - 1) \times (2 - 1) = 1$$

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#### Step 4: Interpret the Result

Using a chi-square distribution table, we compare the calculated chi-square value (1.008) with the critical value at one degree of freedom and a significance level (e.g., 0.05). The critical value, as determined from chi-square distribution tables, is approximately 3.841.

Since  $1.008 < 3.841$ , we fail to reject the null hypothesis. Thus, there is no significant association between gender and product preference in this sample.

### Practical Example - Strong Association

Consider a study investigating the relationship between smoking status (smoker, non-smoker) and the incidence of lung disease (disease, no disease). The researcher collects data from 200 individuals and records the following information:

| Category   | Disease | No Disease | Total |
|------------|---------|------------|-------|
| Smoker     | 50      | 30         | 80    |
| Non-Smoker | 20      | 100        | 120   |
| Total      | 70      | 130        | 200   |

#### Step 1: Calculate Expected Frequencies

Using the formula for expected frequencies:

$$E_{Smoker, Disease} = \frac{(80 \times 70)}{200} = 28 \quad E_{Smoker, Disease} = 200(80 \times 70) = 28$$

$$E_{Smoker, No Disease} = \frac{(80 \times 130)}{200} = 52 \quad E_{Smoker, No Disease} = 200(80 \times 130) = 52$$

$$E_{Non-Smoker, Disease} = \frac{(120 \times 70)}{200} = 42$$

$$E_{Non-Smoker, No Disease} = \frac{(120 \times 130)}{200} = 78$$

**Step 2: Compute Chi-Square Statistic**

$$\chi^2 = \frac{(50-28)^2}{28} + \frac{(30-52)^2}{52} + \frac{(20-42)^2}{42} + \frac{(100-78)^2}{78}$$

$$\chi^2 = \frac{(22)^2}{28} + \frac{(22)^2}{52} + \frac{(22)^2}{42} + \frac{(22)^2}{78}$$

$$\chi^2 = \frac{484}{28} + \frac{484}{52} + \frac{484}{42} + \frac{484}{78}$$

$$\chi^2 = 17.29 + 9.31 + 11.52 + 6.21$$

$$\chi^2 = 44.33$$

**Step 3: Determine Degrees of Freedom**

$$df = (2 - 1) \times (2 - 1) = 1$$

**Step 4: Interpret the Result**

Using a chi-square distribution table, we compare the calculated chi-square value (44.33) with the critical value at one degree of freedom and a significance level (e.g., 0.05), approximately 3.841. Since  $44.33 > 3.841$ , we reject the null hypothesis. This indicates a significant association between smoking status and the incidence of lung disease in this sample.

## Conclusion

The chi-square test is a powerful tool for analyzing the relationship between categorical variables. By comparing observed and expected frequencies, researchers can determine if there is a statistically significant association, providing valuable insights in various fields of study.



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